Mon National College

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GEN-104: Introduction to Research Methodology

Department of Education

A Multidimensional Examination of Malaria Prevalence among Mon National High School and Mon National College Students in Nyi Sar

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Abstract

Malaria remains one of the leading mosquito-borne infectious diseases affecting millions of people worldwide, particularly in tropical and subtropical regions. Despite considerable progress in malaria prevention and treatment, the disease continues to threaten public health in many rural communities, including those in Myanmar. This study examined the prevalence of malaria among students at Mon National High School and Mon National College in Nyi Sar, with the aim of identifying the environmental, socioeconomic, and behavioral factors associated with malaria infection. A mixed-methods research design was employed to obtain both quantitative and qualitative data. Survey questionnaires were administered to students from both institutions to examine demographic characteristics, malaria history, living conditions, and preventive practices. Semi-structured interviews were also conducted to gain deeper insights into students' experiences, awareness, and challenges related to malaria prevention. The findings indicate that malaria prevalence was higher among Mon National High School students than among Mon National College students. Environmental conditions, including proximity to forests and streams, the presence of stagnant water, and limited access to preventive measures, were identified as major contributing factors. In addition, socioeconomic conditions and varying levels of health awareness influenced students' risk of infection. The study concludes that improving health education, strengthening community awareness, and implementing targeted environmental control measures are essential for reducing malaria transmission among students. The findings may assist educational institutions, healthcare providers, and local authorities in developing more effective malaria prevention programs within the Nyi Sar community.

***Keywords*:** malaria prevalence, malaria prevention, Mon National High School, Mon National College, Nyi Sar, students, public health, environmental factors

I. Introduction

Malaria remains one of the most serious infectious diseases worldwide and continues to pose significant challenges to global public health. The disease is caused by *Plasmodium* parasites that are transmitted through the bites of infected female *Anopheles* mosquitoes. Although malaria is both preventable and treatable, it continues to affect millions of people every year, particularly those living in tropical and subtropical regions where climatic conditions support mosquito breeding. According to the World Health Organization (WHO, 2021), an estimated 241 million malaria cases and 627,000 malaria-related deaths were reported globally in 2020. The disease continues to place a heavy burden on developing countries by affecting public health, education, and economic productivity.

Myanmar remains one of the countries where malaria continues to be a public health concern, particularly in rural and forested regions. Communities living near forests, rivers, and stagnant water sources are often exposed to higher mosquito populations, increasing the risk of malaria transmission. Although national malaria control programs have reduced infection rates in many areas, certain communities continue to experience seasonal outbreaks because of environmental conditions, limited healthcare access, and insufficient awareness of preventive practices (WHO, 2021).

Nyi Sar is a rural area where malaria remains a common health issue. Based on local observations, students attending Mon National High School appear to experience malaria more frequently than students studying at Mon National College. This difference has raised concerns among educators and community members regarding the environmental and social factors that may contribute to higher infection rates. Students living close to forests or streams may be exposed to more mosquito bites, while differences in housing conditions, health awareness, and access to preventive measures may further influence their vulnerability to malaria.

Environmental conditions play an important role in malaria transmission. Mosquitoes reproduce successfully in warm, humid environments where stagnant water provides suitable breeding sites. Forested areas with limited sunlight and high moisture also create favorable habitats for mosquito populations (Githeko et al., 2000). In addition, malaria transmission often occurs in localized hotspots where environmental conditions allow mosquitoes to survive throughout the year (Bousema et al., 2012). These environmental characteristics are commonly found in rural areas such as Nyi Sar, making malaria prevention particularly challenging.

Socioeconomic factors also influence malaria prevalence. Poor housing conditions, low household income, limited healthcare access, and inadequate knowledge about malaria prevention increase individuals' risk of infection (Tusting et al., 2013). Students from economically disadvantaged families may have limited access to insecticide-treated mosquito nets, protective clothing, or timely medical treatment. Consequently, differences in socioeconomic conditions may partly explain variations in malaria prevalence between student populations.

Education plays a vital role in malaria prevention. Students who understand the symptoms of malaria and the importance of preventive practices are more likely to seek early treatment and protect themselves against mosquito bites. Public awareness campaigns, school-based health education, and community participation have been recognized as effective approaches for reducing malaria transmission (Fox & Chaplin, 2022). However, little research has specifically compared malaria prevalence between Mon National High School students and Mon National College students within the Nyi Sar community.

Therefore, this result aims to examine the prevalence of malaria among students at both educational institutions and identify the environmental, socioeconomic, and behavioral factors associated with malaria infection. By understanding the causes of differing infection rates, this research seeks to provide practical recommendations that can support educational institutions, healthcare providers, and local communities in strengthening malaria prevention programs and improving students' health and well-being.

II. Literature Review

Malaria remains one of the world's most significant public health challenges despite decades of prevention and control efforts. The disease is caused by *Plasmodium* parasites, which are transmitted through the bites of infected female *Anopheles* mosquitoes. Although considerable progress has been made in reducing malaria-related deaths through improved diagnosis, treatment, and vector control, malaria continues to affect millions of people, particularly in developing countries. According to the World Health Organization (WHO, 2021), malaria remains a major health burden in tropical and subtropical regions, where environmental conditions support year-round mosquito breeding. Children, adolescents, and individuals living in rural communities continue to face the highest risk of infection because of frequent exposure to mosquito habitats and limited access to healthcare services.

Environmental conditions are widely recognized as one of the strongest determinants of malaria transmission. Mosquitoes breed in stagnant water such as ponds, streams, irrigation canals, and uncovered water containers. Warm temperatures, high humidity, and dense vegetation further increase mosquito survival and reproduction. Githeko et al. (2000) explained that climate and environmental changes directly influence mosquito populations and the spread of vector-borne diseases. Likewise, Bousema et al. (2012) argued that malaria transmission is often concentrated in specific geographical "hotspots," where environmental conditions create favorable breeding sites for mosquitoes. In rural communities like Nyi Sar, where forests, streams, and moist environments are common, students may experience greater exposure to mosquito bites than those living in more urbanized areas.

In addition to environmental factors, socioeconomic conditions significantly influence malaria prevalence. Poor housing, limited financial resources, and inadequate access to healthcare increase people's vulnerability to infection. Families with low incomes may struggle to purchase insecticide-treated mosquito nets or seek timely medical treatment when symptoms appear. Furthermore, houses without proper window screens or protective barriers provide mosquitoes with easier access to residents during the evening. A systematic review by Tusting et al. (2013) found that socioeconomic development plays a critical role in reducing malaria transmission because improved housing, sanitation, and healthcare access significantly lower infection rates. These findings suggest that socioeconomic inequalities may partly explain differences in malaria prevalence among students from different educational institutions.

Health education and public awareness are also essential components of malaria prevention. Students who understand how malaria is transmitted are more likely to adopt preventive behaviors, including sleeping under mosquito nets, eliminating mosquito breeding sites, wearing protective clothing, and seeking medical attention when symptoms occur. According to Fox and

Chaplin (2022), increasing public awareness is particularly important for vulnerable populations, including children and adolescents, because early recognition of symptoms can reduce severe illness and prevent complications. Similarly, the World Health Organization (2021) emphasized that community education, early diagnosis, and prompt treatment remain among the most effective strategies for reducing malaria-related morbidity and mortality.

Preventive interventions have proven highly effective when consistently implemented within communities. One of the most successful malaria prevention strategies is the use of insecticide-treated mosquito nets (ITNs). Killeen et al. (2007)reported that widespread use of ITNs substantially reduces mosquito bites and lowers malaria infection rates, particularly among children and young adults. Additional preventive measures, including indoor residual spraying, environmental sanitation, and community participation in eliminating mosquito breeding sites, further contribute to reducing malaria transmission. These strategies demonstrate that malaria prevention requires cooperation between individuals, schools, healthcare providers, and local authorities.

Previous studies have also highlighted the importance of demographic and geographical characteristics in understanding malaria prevalence. Snow et al. (2005) found that malaria distribution varies considerably across regions because of differences in climate, population density, healthcare infrastructure, and environmental conditions. Rural communities located near forests and water sources generally experience higher transmission rates than urban communities. These findings are particularly relevant to the present study because both Mon National High School and Mon National College are located within a rural environment, yet anecdotal evidence suggests that malaria prevalence differs between the two student populations.

Although considerable research has examined malaria transmission globally, limited studies have specifically investigated malaria prevalence among students in educational settings within Myanmar. Existing literature primarily focuses on national or regional malaria trends rather than comparing students from different educational institutions. Consequently, little is known about how environmental exposure, socioeconomic conditions, and students' knowledge collectively influence malaria prevalence among Mon National High School and Mon National College students in Nyi Sar. This study seeks to address this gap by providing a multidimensional examination of malaria prevalence within these two institutions. By integrating environmental, socioeconomic, and behavioral perspectives, the research aims to contribute evidence that can inform future malaria prevention strategies and improve student health within the local community.

III. Methodology

Research Design

This study employed a **mixed-methods research design**, combining quantitative and qualitative approaches to obtain a comprehensive understanding of malaria prevalence among students at Mon National High School and Mon National College in Nyi Sar. The quantitative component measured the prevalence of malaria and examined the relationship between environmental, socioeconomic, and behavioral factors. The qualitative component provided a deeper understanding of students' experiences, awareness, and perceptions regarding malaria prevention and treatment. Using both approaches allowed the researcher to validate findings through multiple sources of evidence and develop a more complete understanding of the research problem.

Participants and Sampling

The study was conducted among students from Mon National High School and Mon National College in Nyi Sar. A total of 40 students participated in the quantitative survey, including 20 students from Mon National High School and 20 students from Mon National College. Participants were selected using stratified random sampling to ensure representation from both institutions.

To complement the survey findings, eight students (four from each institution) participated in semi-structured interviews. The interview participants were selected using purposive sampling because they had either experienced malaria or possessed relevant knowledge about malaria prevention within their communities.

Data Collection

Quantitative data were collected using a structured questionnaire consisting of four sections: demographic information, malaria infection history, environmental and socioeconomic conditions, and malaria prevention practices. The questionnaire included multiple-choice questions and five-point Likert-scale items to measure students' awareness and preventive behaviors.

Qualitative data were collected through semi-structured interviews lasting approximately 20–30 minutes. The interviews explored students' experiences with malaria, knowledge of disease transmission, prevention practices, challenges in accessing healthcare, and recommendations for improving malaria awareness within their communities.

Data Analysis

The quantitative data were analyzed using descriptive statistics, including frequencies, percentages, and mean scores, to summarize participants' responses. Comparative analyses were conducted to identify differences between students from Mon National High School and Mon National College regarding malaria prevalence and associated risk factors.

The qualitative interview data were analyzed using thematic analysis. Interview transcripts were carefully reviewed, coded, and grouped into common themes, including environmental exposure, health awareness, preventive practices, and barriers to malaria prevention. The integration of quantitative and qualitative findings strengthened the credibility and validity of the study.

Ethical Considerations

Ethical principles were carefully observed throughout the research process. Participation was voluntary, and informed consent was obtained from all participants before data collection. Students were informed about the purpose of the study and assured that their responses would remain confidential and anonymous. Personal information was not included in the final report, and participants were free to withdraw from the study at any stage without any negative consequences.

IV. Results

4.1 Demographic Characteristics

A total of **40 students** participated in the survey, including **20 students from Mon National High School** and **20 students from Mon National College**. The participants ranged in age from **16 to 22 years**, with both male and female students represented. Most participants had lived in Nyi Sar for more than five years, indicating that they were familiar with the local environment and malaria situation.

Among the respondents, **60%** reported having experienced malaria at least once during their lifetime. However, malaria prevalence differed considerably between the two institutions. Approximately **75% of Mon National High School students** reported previous malaria infection compared with **45% of Mon National College students**. This finding suggests that high school students were more vulnerable to malaria than college students.

4.2 Environmental Factors

The survey results indicated that environmental conditions were strongly associated with malaria prevalence. Nearly **70%**of students who had experienced malaria reported living near forests, streams, or stagnant water sources. These environments provide favorable breeding habitats for *Anopheles* mosquitoes, increasing exposure to mosquito bites.

Furthermore, **65%** of participants stated that mosquitoes were abundant around their homes during the rainy season. Students living in forested areas also reported spending more time outdoors during the evening, which further increased their risk of mosquito exposure.

4.3 Socioeconomic Factors

Socioeconomic conditions also appeared to influence malaria prevalence. About **55%** of respondents indicated that their families had limited financial resources to purchase mosquito repellents, insecticide-treated nets, or other preventive materials. Additionally, several students reported that healthcare facilities were located far from their villages, making early diagnosis and treatment more difficult.

Students from Mon National College generally reported better access to health information and preventive resources than students from Mon National High School. This difference may partly explain the lower malaria prevalence among college students.

4.4 Malaria Awareness and Prevention Practices

The findings revealed that most participants understood that malaria is transmitted through mosquito bites. Approximately **85%** correctly identified mosquitoes as the primary source of infection. However, only **58%** reported consistently sleeping under insecticide-treated mosquito nets, while **42%** admitted that they used mosquito nets only occasionally or not at all.

Interview participants explained that hot weather, damaged mosquito nets, and limited availability often discouraged regular use. Several students also believed that malaria was unavoidable during the rainy season because mosquitoes were widespread throughout the community.

4.5 Interview Findings

The interviews provided a deeper understanding of students' experiences with malaria. Four major themes emerged from the analysis.

Environmental Exposure: Most interviewees believed that living near forests and streams increased their chances of contracting malaria. One participant explained, *"There are many mosquitoes around our house because we live close to the forest. During the rainy season, mosquitoes become even more common."*

Health Awareness: Although students recognized the symptoms of malaria, many admitted that they delayed seeking medical treatment because they initially believed the symptoms were caused by common fever or seasonal illness.

Challenges in Prevention: Several participants reported financial difficulties in purchasing mosquito repellents and replacing damaged mosquito nets. Others mentioned that community awareness campaigns were conducted only occasionally.

Recommendations: Participants suggested increasing health education programs in schools, improving access to mosquito nets, conducting regular environmental clean-up campaigns, and strengthening cooperation between schools and local healthcare providers to reduce malaria transmission.

Overall, the study suggests that environmental conditions, socioeconomic status, and preventive behaviors collectively contribute to the higher prevalence of malaria among Mon National High School students compared with Mon National College students. These findings provide the foundation for the discussion presented in the following section.

V. Discussion

This study examined the prevalence of malaria among students at Mon National High School and Mon National College in Nyi Sar and explored the environmental, socioeconomic, and behavioral factors associated with malaria infection. The findings indicate that malaria prevalence was higher among Mon National High School students than among Mon National College students. The results suggest that environmental exposure, living conditions, health awareness, and preventive practices collectively influenced students' vulnerability to malaria.

One of the most significant findings of this study is that students living near forests, streams, and stagnant water were more likely to experience malaria infection. Most participants who reported previous malaria infection lived in areas surrounded by dense vegetation and water sources, which provide ideal breeding environments for *Anopheles* mosquitoes. This finding is consistent with Githeko et al. (2000), who reported that environmental conditions such as temperature, humidity, and standing water strongly influence mosquito populations and the transmission of malaria. Likewise, Bousema et al. (2012) found that malaria transmission is often concentrated in geographical hotspots where favorable environmental conditions allow mosquitoes to survive and reproduce throughout the year. Since many students in Nyi Sar live close to forests and streams, their daily exposure to mosquitoes naturally increases the risk of infection.

The study also found that socioeconomic conditions influenced malaria prevalence. Students from families with limited financial resources reported greater difficulty obtaining insecticide-treated mosquito nets, mosquito repellents, and timely medical treatment. Some participants also explained that healthcare facilities were located far from their villages, making it difficult to receive early diagnosis and treatment. These findings support the study by Tusting et al. (2013), which concluded that improved socioeconomic conditions, including better housing, healthcare access, and living standards, contribute significantly to reducing malaria transmission. Therefore, improving community infrastructure and healthcare accessibility may reduce malaria risk among students.

Another important finding concerns students' awareness and preventive practices. Although most participants correctly understood that malaria is transmitted through mosquito bites, many admitted that they did not consistently use mosquito nets or other protective measures. Some students reported that damaged mosquito nets, hot weather, or limited availability discouraged regular use. Others delayed seeking medical treatment because they believed their symptoms were caused by ordinary fever rather than malaria. These findings indicate that knowledge alone does not always result in appropriate preventive behavior. According to the World Health Organization (2021), continuous health education and early diagnosis are essential for reducing malaria-related illness and preventing severe complications. Similarly, Fox and Chaplin (2022) emphasized that increasing community awareness can improve preventive practices and encourage individuals to seek treatment before the disease becomes severe.

The interview findings further demonstrated the importance of community participation in malaria prevention. Participants recommended strengthening school-based health education, organizing environmental clean-up campaigns, and improving access to mosquito nets within local communities. These suggestions are consistent with Killeen et al. (2007), who reported that widespread use of insecticide-treated mosquito nets and community-based prevention programs significantly reduce malaria infection rates. Schools can therefore play an important role by integrating malaria education into health programs and encouraging students to participate in community prevention activities.

Although Mon National High School and Mon National College are located within the same geographical region, differences in students' living environments, preventive practices, and access to health information may explain the variation in malaria prevalence observed in this study. College students generally demonstrated greater awareness of malaria prevention and reported better access to educational resources than high school students. This suggests that education itself may contribute to healthier behaviors and improved decision-making regarding disease prevention.

Practical Implications

The findings of this study have several practical implications for educational institutions and public health organizations. First, schools should provide regular health education programs that teach students how malaria is transmitted, how to recognize early symptoms, and how to protect themselves from mosquito bites. Second, local authorities should strengthen environmental management by removing stagnant water, improving sanitation, and conducting mosquito control activities, especially during the rainy season. Third, greater collaboration between schools, healthcare providers, and community leaders can improve awareness campaigns and ensure that preventive resources such as insecticide-treated mosquito nets are available to students. Together, these efforts can contribute to reducing malaria transmission and promoting healthier learning environments in Nyi Sar.

VI. Conclusion

Malaria continues to be a significant public health concern in rural communities such as Nyi Sar, where environmental conditions and socioeconomic challenges contribute to ongoing transmission. This study examined the prevalence of malaria among students at Mon National High School and Mon National College and identified the factors associated with differences in infection rates between the two institutions.

The findings indicate that Mon National High School students experienced a higher prevalence of malaria than Mon National College students. Environmental factors, including proximity to forests, streams, and stagnant water, were identified as major contributors to increased mosquito exposure. In addition, socioeconomic conditions such as limited financial resources, restricted access to healthcare services, and inconsistent use of preventive measures further increased students' vulnerability to malaria. Although most participants demonstrated a basic understanding of malaria transmission, preventive practices were not consistently applied, highlighting the need for continued health education and community support.

The study also emphasizes the importance of collaboration among educational institutions, healthcare providers, and local communities. School-based health education, improved environmental sanitation, increased access to insecticide-treated mosquito nets, and early medical intervention can collectively reduce malaria transmission among students. These preventive strategies not only protect students' health but also contribute to improved school attendance, academic performance, and overall well-being.

Despite providing valuable insights, this study has several limitations. The sample size was relatively small and included students from only two educational institutions within Nyi Sar. Consequently, the findings may not fully represent malaria prevalence in other communities. Future research should involve larger and more diverse samples, incorporate clinical data where possible, and examine additional factors such as seasonal variations, family health practices, and community-based intervention programs.

Overall, this finding contributes to a better understanding of malaria prevalence among students in Nyi Sar by demonstrating that environmental exposure, socioeconomic conditions, and preventive behaviors collectively influence the risk of malaria infection. The findings provide practical recommendations that may assist schools, healthcare professionals, and policymakers in developing targeted malaria prevention strategies that promote healthier communities and improve students' quality of life.

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